

- [TradePilot – Full Detailed Report \(June 2026\)](#)
- [1. Executive Summary](#)
- [2. Root-Cause Investigation – Why We Were Losing](#)
 - [2.1 The edge is execution discipline, not ML/AI](#)
 - [2.2 Complexity diluted the edge](#)
 - [2.3 Forensic trade-level findings](#)
 - [2.4 The decisive finding – the SHORT book is the net bleed](#)
 - [2.5 v5 classic \(“the old tree”\) dominance](#)
 - [2.6 The OLD 5-tree A/B – a leverage mirage hiding a real gem](#)
 - [2.7 Watchdog validation – entry quality, not volume](#)
- [3. Debugs & Data-Integrity Fixes](#)
 - [3.1 The 06-26 all-day data stall](#)
 - [3.2 Conviction not stored on closed trades](#)
- [4. Engine Roster Consolidation](#)
- [5. Competitive & Strategy Research](#)
 - [5.1 Competitive landscape \(105-agent deep-research, adversarially verified\)](#)
 - [5.2 Autonomy benchmark – T3](#)
 - [5.3 Tier-2 – Indian retail-algo market](#)
 - [5.4 Tier-4 – institutional AI funds](#)
 - [5.5 SEBI compliance \(the answer to “are we doing anything wrong?”: **No**\)](#)
 - [5.6 3-pillar strategy research – the convergence](#)
- [6. The Red-Day / Regime-Flip Arc \(06-30\) – Four Data Validations](#)
 - [6.0 The live diagnosis](#)
 - [6.1 The two structural gaps \(✅ confirmed in code\)](#)
 - [6.2 Validation 1 – “no flips after 13:30” is REFUTED](#)
 - [6.3 Validation 2 – “short=red, long=green” is REFUTED](#)
 - [6.4 Validation 3 – tilt magnitude & trigger](#)
 - [6.5 Validation 4 – fixed ratio vs dynamic allocation](#)
 - [6.6 The 10% cap – clarification](#)
- [7. Features & Tools Built](#)
 - [7.1 Red-day watchdog – `scripts/red-day-watchdog.py`](#)
 - [7.2 Daily engine compare – `scripts/engine-compare.py` + `launchd`](#)
 - [7.3 Decision dashboard – `docs/decision-dashboard.html` + `Flask` /`decisions` + `pageswitch`](#)
 - [7.4 yfinance per-process cache fix – `data\_nse.py` + `missed-opportunities-watchdog.py`](#)
 - [7.5 Conviction logging – all engines](#)
 - [7.6 `v5\_flip` fast-regime-flip shadow – `scripts/v5\_flip-paper-trade.py` + `wiring`](#)
- [8. Roadmap \(all tracks\)](#)
- [9. Learnings Index \(DevPilot store\)](#)
- [10. Open Questions & Next Steps](#)

TradePilot – Full Detailed Report (June 2026)

The complete record: every finding, every debug, every feature, every validation – with the data, evidence, code locations, and reasoning behind each.

Project	TradePilot (autonomous agentic trading, NSE intraday)
Document	Full detailed report – June 2026 work arc
Version	v1.0.0
Period covered	2026-06-22 → 2026-06-30
Method	14+ research/forensic agents, ~7.5M tokens of deep-research, Sarathi-disciplined data validation
Status	Complete
Author	Soumya Swain
Email	soumya@sidewall.in

1. Executive Summary

This report documents an eight-day investigation that began with a single question – “*why are we losing money?*” – and ended with a data-grounded fix shipped into testing. The arc:

1. **Root cause** – The edge was never the AI/ML; it is **execution discipline**. Three months of reactive complexity (NIFTY-50→200, long+short, 4 pools, a 1,735-tree ML model, late-entry rescore loop) **diluted** a profitable engine. Win-rate fell **82% → ~48%** as trades/day rose **17 → 45**. The frozen original (`v5_classic`) quietly out-earns the live rebuild.
2. **Competitive & strategy** – TradePilot has **no direct competitor** (no Indian platform is agentic; global peers are mostly research prototypes), sits at autonomy level **T3** (the level credible real funds run at), and the academic literature **validates** its core thesis (LLM/ML alpha is largely lookahead-bias illusion). A 3-pillar strategy (profitability, UX, platform) showed all three converge on one artifact: a **simple, explainable, clean-execution decision pipeline**.
3. **Red-day regime work** – A live red day exposed that the engine **does not adapt its long/short mix to the tape**. Four successive data validations (each correcting a coarse assumption of mine) reshaped the fix from “flip the book short” into “**tilt the ratio, decide direction per-stock, with a risk cap.**”
4. **Builds shipped** – Conviction logging across **all** engines (closing a “store-everything” gap), and a `v5_flip` fast-regime-flip shadow added to the A/B rotation. Plus three infra/diagnostic tools (red-day watchdog, daily compare, decision dashboard) and a data-stall root fix.

Bottom line: the engine is not broken – its *execution* degraded and it *doesn't read the tape*. Both are fixable and now instrumented. The path is sequenced: **prove profitability → go live cheaply (own capital) → productize into an agentic platform only if chosen.**

2. Root-Cause Investigation – Why We Were Losing

2.1 The edge is execution discipline, not ML/AI

The original profitable engine (April) ran NIFTY-50, top-5 long-only, a tight +1.5% / –0.75% bracket, flat at 15:15 – and hit an **82% win rate**. Decisive evidence the *models* were not the source of that edge:

- On the 82%-WR day, the average **winner's composite score (66.9) ≈ the average loser's score (69.2)**. The score was not separating winners from losers.
- The ML information coefficient was **0.006** – statistically indistinguishable from zero. ML was later weighted to 0 with no measured degradation.

The money came from the *disciplined bracket + concentration + end-of-day flat exit*, not the intelligence.

2.2 Complexity diluted the edge

DIMENSION	ORIGINAL (APRIL)	CURRENT (JUNE)	DIRECTION
Universe	NIFTY-50	NIFTY-200	4× wider
Direction	Long-only	Long + short + flip	More exposure
Positions	5	20 (4 pools)	4× more
Trades/day	17	~45-57	2.6-3.4× more
Bracket	+1.5% / -0.75%	+2.0% / -1.5%	Wider stops
Win rate	82%	44-53%	Collapsed

Rising trade count with falling win rate is the mathematical signature of **over-diversification + over-trading** regressing a selective alpha toward a coin flip. The complexity cascade was *reactive* – each layer (regime detector, kill switches, shorts, pools, FLAT_FORCE_EXIT) was bolted on after a single bad day (notably v4's -₹30,816 loss on Apr-9, a bear day with no regime filter), not because the simple engine was failing.

2.3 Forensic trade-level findings

FINDING	DETAIL	VERDICT
FLAT_FORCE_EXIT exonerated	Trade-level audit of 06-17 & 06-22 (the two worst v5-specific days): on the symbols it flattened, it ended +₹30 and +₹299 better than the frozen original holding them	Innocent – it helped
Late entry is the day-to-day bleed	v5 entered the <i>same names, same direction</i> 30-60 min later than v5_classic on 16/35 (06-17) and 21/38 (06-22) shared positions → winners booked as STOPLOSS/TIME_EXIT not TARGET. Cost: -₹1,969 (06-17), -₹1,092 (06-22)	The real mechanism
Over-trading = cost drag	8 days: gross +₹4,691 , round-trip cost – ₹7,153 → net -₹2,462 . Excess churn from 57/day vs original 17/day = ₹5,046 = 205% of net . At 17/day cadence the same edge nets +₹2,584 (winning)	Cost flips a winning book to a loss
Under-deployed, fragmented	Avg position ₹13,130 (~10× smaller than the 15%-of-pool rule implies); only ~₹260k working on a ₹10L book	Capital barely used

2.4 The decisive finding – the SHORT book is the net bleed

Across 454 trades (06-16 → 06-25):

DIRECTION	TRADES	NET P&L	WIN RATE	TARGET HIT
LONG	213	+₹1,149	41.8%	8.5%
SHORT	241	-₹3,611	34.4%	4.1%
Total	454	-₹2,462	37.9%	6.2%

If the short book were merely flat, v5 would be net positive. Shorts hit TARGET only **4.1%** of the time (longs 8.5%) and die on **STOPLOSS (-₹9,107, the single biggest loss bucket)** – classic shorting-into-an-up-drifting-tape. (Note: this finding is refined later in §6.2 – shorts are not categorically bad; mis-timed shorts on the wrong tape are.)

2.5 v5_classic ("the old tree") dominance

The frozen original v5 (git [236d6e4](#), pre-Rust, pre-safeguards) has out-performed the live rebuild over 45 sessions (Apr 20 → Jun 29):

METRIC	V5_CLASSIC	V5 (LIVE)
Cumulative (45 sessions)	+₹111,803	+₹116,357
Green / red days	28G / 14R (62%)	fewer
Beat the other, day-by-day	won 29/43 (67%)	won 33%
Avg/day	+₹2,485	–

v5's apparent ₹4,554 lead is a mirage – it comes entirely from **two April outlier days**: 04-22 (v5 +₹44,612 vs classic +₹10,291, a ₹34,321 gap) and 04-23 (+₹15,352 gap) = ₹49,673. Strip those and v5_classic wins the remaining 43 sessions decisively. This is the *identical* outlier-fragility that retired v4.

The trend is decisive and accelerating:

MONTH	V5_CLASSIC	V5 (LIVE)	WINNER
April (early A/B)	+₹39,245 (8G/1R)	+₹69,342	v5 – <i>only via the 2 outlier days</i>
May	+₹59,667 (won 13/17)	+₹46,325	classic
June	+₹12,891 (11G/5R)	+₹690	classic, 18×

Why it wins: clean early-entry execution + holds to TIME_EXIT, with **none** of the late-entry lag, FLAT_FORCE_EXIT, or 57-trade churn the rebuild carries. Same signals, same universe, same stops – the original *executes* them cleanly.

2.6 The OLD 5-tree A/B – a leverage mirage hiding a real gem

The dashboard card “v4 model – 5-tree challenger vs live 1,735-tree” (sibling project [tradepilot-oldengine-ab](#)) shows the OLD 5-tree model ([best_iteration: 5](#) vs the live model's **1,735**), long-only, printing **+₹122,003 over 14 sessions (71% green)** vs live v5's -₹6,023. Skeptic checks deflate the headline:

CHECK	FINDING	EFFECT
Leverage	06-29 deployed ₹3,130,953 = 313% of the ₹10L book (139 positions) vs v5's ~35%	~9× more capital – bigger rupees, not better alpha
Gross, not net	<code>realized_pnl</code> only, no cost field (v5's is net of 12bps)	Costs would erase a large chunk
Outlier-driven	Top 4 days = ₹88,303 = 72% of total; the other 10 days net ~₹34k	Same fragility that retired v4

The real, leverage-independent gem: a 5-boosting-round model **out-hits** the 1,735-round live model. This is hard internal evidence the **live ML model is overfit** (textbook: complex models memorize noise; simple models generalize) – and the fix it implies (“shrink the model drastically”) is cheap and testable.

2.7 Watchdog validation – entry quality, not volume

Independent telemetry (06-16 → 06-25) cross-checked the claims:

- Only **6.2% of 454 exits hit TARGET**; the book held **3.6× more losers than winners** at any snapshot.
- **Win-rate drives P&L, not trade volume:** 06-25 did 67 trades for **+₹5,014 (61% WR)**; 06-24 did 62 trades for **–₹2,044 (31% WR)**. Near-identical volume, opposite result.
- **“What right looks like”:** 06-19 (BEAR regime) – 45 trades, **zero** SIGNAL_FLIP/FLAT churn, **+₹1,602**.

3. Debugs & Data-Integrity Fixes

3.1 The 06-26 all-day data stall

Symptom: On 06-26 every engine made ~5 trades all day, ₹0 P&L, the missed-ops watchdog flatlined at 10:31, engines alive but starved.

Root cause (✓ verified): yfinance 1.2.0 keeps a shared SQLite cache (`~/Library/Caches/py-yfinance/tkr-tz.db`). All engine processes + dashboard + watchdogs hit the *same* file concurrently → `OperationalError: unable to open database file` → every download returns `None` → no prices → no signals → no trades. The `-wal` / `-shm` files present at the time are the smoking gun of concurrent SQLite access. Not a sleep (heartbeats ran) or crash (nothing restarted).

Fix: Per-process cache isolation – `yf.set_tz_cache_location()` keyed by `ENGINE_NAME` /PID – in `prototype/v4/data_nse.get_yfinance()` (covers all engines + dashboard) and `scripts/missed-opportunities-watchdog.py` (calls yfinance directly). Cleared the corrupt shared db. **Verified:** live `RELIANCE.NS` download works post-fix; held under real multi-engine load on 06-29 (all 4 engines normal volume, no repeat).

3.2 Conviction not stored on closed trades

Symptom: `score` / `reasons` could not be correlated to P&L from history → the dynamic-allocation question (§6.4) was unprovable.

Root cause (✓ verified): The entry conviction is captured (`score=33.0` , `direction` , `reasons` all live on the open position), but `close_position` rebuilt the closed record from a **hardcoded subset of fields** and dropped them. The data was never missing – it was *thrown away at the last step*. Against the standing “store everything” instruction.

Fix: Carry `score` , `direction` , `reasons` , `sl_price` , `target_price` , `entry_date` , `trailing_activated` into the closed record across **all** engines: the v5 family (`v5` , `v5_long` , `v5_cut`) via the shared `close_position` , plus the standalone `v5_classic` and `v7_regime` scripts. **Verified:** compile clean, smoke gate PASS (exit 0), fields present in the close path. Applies at next launch; running processes untouched. From the next session every closed trade is fully instrumented; ~5 sessions yields conviction→P&L data.

4. Engine Roster Consolidation

6 → 4 lean roster (06-26 audit), then → 5 with the new shadows. Retirements are *commented out* (state preserved), never deleted.

ENGINE	ROLE	STATUS
v5	Live baseline (with shorts) – the thing to beat	Active
v5_classic	Frozen April original – the durable benchmark	Active
v5_long	RC-1: long-only NIFTY-200 (shorts disabled via <code>SHORT_REQ_MAX_SCORE=-1</code>)	Active
v5_cut	ML-removed + faster wrong-way cut + tighter short-gate + ~450 universe	Active
v5_flip	Fast intraday regime-flip (new, \$7.6)	Active (new)
v5_neml	Redundant – <code>ml_score=0</code> already global, so it ran v5 twice	Retired
v5_apr	Tracked v5 within +₹78 over 9 days – no information value	Retired
v7_regime	Beat v5 by only +₹1,684/9d; daily-gate WFO showed no edge (DSR 0.12)	Retired

Consistency enforced across `launch-market.sh`, `crash-watchdog.sh`, and `engine-compare.py` (all show the same 5 active engines).

5. Competitive & Strategy Research

5.1 Competitive landscape (105-agent deep-research, adversarially verified)

TradePilot has no direct head-to-head competitor. Genuinely agentic trading exists but is ~95% open-source research frameworks / academic prototypes: TradingAgents (4 analysts + bull/bear debate), FinMem (layered memory), TradingGroup (5 agents + self-reflection), The Self-Driving Portfolio (~50 agents + a meta-agent that rewrites its own code), AI-Trader, FinWorld, FinRL. The one commercial Tier-1 peer, **Standard Signal (YC S26)**, makes live-trading + Sharpe>3 claims that **failed verification** (unaudited, solo founder; the “first” claim is false – Aidyia/Sentient predate it by a decade).

Reality-check (3-0 verified): LLM trading alpha is **largely lookahead-bias / memorization** – it collapses out-of-sample (one cited case: +20.73% in-sample → -1.04% post-cutoff). Removing LLM features dropped Sharpe only 1.40→1.14. This **independently validates** the decision to zero our ML (IC 0.006): the edge living in *execution discipline* is the more defensible position.

5.2 Autonomy benchmark – T3

No trading-specific L0-L5 standard exists; derived from the Knight Columbia agent-autonomy paper (arXiv 2506.12469): **TradePilot = T3 (Conditional Autonomy)** – AI runs the entire intraday loop (research → regime → score → signal → size → execute → EOD) with no human in the loop; humans retain veto only at the strategy layer. T4 would require self-modifying strategy params without sign-off. T3 is the level credible real funds actually operate at.

5.3 Tier-2 – Indian retail-algo market

Zero Indian platforms are genuinely agentic. Every major one – Streak, Tradetron, AlgoTest, QuantMan, Sensibull, AlgoBulls – is a rule-based strategy builder where the user defines all logic. The closest (AlgoBulls Phoenix Copilot) just translates “buy when RSI<30” into code. TradePilot’s autonomy is a **category gap**, not a feature gap. Incumbents’ moats are elsewhere: live execution (Tradetron 1.5M trades/mo, 60+ brokers), real users (Sensibull 1M+, AlgoTest ~25k), 2+ years of SEBI compliance work, polished UX, and ₹300-1,300/mo pricing the market is trained on.

5.4 Tier-4 – institutional AI funds

Every fund with real money keeps a human in the strategy loop: Renaissance & Two Sigma frame AI as a research/signal tool, not the decision-maker. The “fully autonomous” funds (Lumenai, Altbridge, Standard Signal) are pre-launch / proof-of-concept / self-reported.

Numerai is the closest verified AI-native (JPMorgan committed up to \$500M; ~25% net 2024) – but it’s crowdsourced ensemble + systematic execution, not autonomous agents.

5.5 SEBI compliance (the answer to “are we doing anything wrong?”: No)

SEBI’s framework (Feb-2025 circular, fully in force **April 1, 2026**) regulates *orders on exchanges via broker APIs*. TradePilot places none.

TIER	TRIGGER	OBLIGATIONS	BURDEN
Paper (now)	Simulation only	None	Zero
Live, own capital	Kite order API	Static IP + whitelist · kill-switch · 5-yr order audit log · Kite algo-TOS check · OPS profiling (we run ~ 0.0024 OPS avg, peak <1 – far under the 10-OPS line) · family-account carve-out only	Light
Offer to others	Product	SEBI Research Analyst licence (black-box) · NSE/BSE empanelment · ISO 27001 · biannual CERT-In VAPT · per-strategy algo-ID · grievance redressal	Heavy (6-12 mo)

Running the algo for anyone outside the family (self/spouse/dependent parents/children) instantly triggers the heavy “algo provider” tier.

5.6 3-pillar strategy research – the convergence

All three pillars point at one artifact:

- **Profitability (5 ranked levers):** ① execute-at-open / kill the rescore loop (highest, free; Man Group: alpha decays in minutes) · ② collapse the ML to ≤20 trees + purged walk-forward (Bailey/Lopez-de-Prado: ~4.75 backtests = Sharpe 1.0 by chance) · ③ ATR-sizing at ≥60% deployment (fractional Kelly; underbetting destroys returns) · ④ binary regime gate on *direction* (SMA + VIX-rank, no grid-search) · ⑤ retrain on the correct label, P(target before stop).
- **UX:** the user is a **supervisor, not a pilot** – single status surface, plain-language decision log, progressive-delegation onboarding, Telegram check-not-operate, dual-mode UI.
- **Platform:** “**BYOB White-Box Agent**” – runs in the user’s own broker account, zero platform capital (no PMS licence), white-box (no RA licence). 9-role agent DAG, train/eval-namespaces separation (structural anti-overfit).

The spine: the explainable decision log is simultaneously the profitability validation, the UX trust surface, the SEBI compliance audit trail, *and* the platform’s audit layer. One build, four payoffs. The agentic-DAG architecture is the *same* investment as the overfitting fix.

6. The Red-Day / Regime-Flip Arc (06-30) – Four Data Validations

This is the most rigorous sequence of the arc. The `/rules` (Sarathi) discipline was explicitly applied: **re-verify instead of defend; show evidence; confidence-label every claim**. The data corrected a coarse assumption of mine **four times**.

6.0 The live diagnosis

At 09:48 on a red day (NIFTY -0.73% intraday), per-engine attribution was unambiguous:

ENGINE	NET	LONGS	SHORTS
v5	-1,214	-1,671 (17t)	+623 (10t)
v5_classic	-1,753	-2,376 (16t)	+623 (10t)
v5_long	-1,238	-1,142 (23t)	0 (no shorts)
v5_cut	-1,082	-585 (17t)	-371 (10t)

The longs were the entire bleed; the shorts were green. The engines were long-heavy (16-23 longs vs 10 shorts) on a falling tape.

6.1 The two structural gaps (✅ confirmed in code)

- **Gap 1 – regime frozen at launch.** `detect_regime()` runs once at premarket (`v5-paper-trade.py:379`); `rescore_and_redeploy` reuses `state["regime"]` (`:698`) and never re-detects intraday.
- **Gap 2 – slow inputs.** `regime_detector.py` votes on 6 daily aggregates (50/200-DMA, 5-day momentum, daily VIX, FII/DII, advance-decline). A red open barely moves a 50-day average → it reads SIDEWAYS/BULL → deploys the 15-long/5-short split. The engine is **structurally blind to a red open**.

6.2 Validation 1 – “no flips after 13:30” is REFUTED

I had proposed going defensive after 13:30. The data (INTRADAY trades, all sessions since April), by **entry** time-of-day:

V5 (50 SESSIONS)	NET P&L
Entered after 1:00pm	+₹30,515 (486t)
Entered before 1:00pm	-₹2,724 (723t)
Entered after 1:30pm	+₹20,209 (would have been thrown away)

Best entry hour = **13:00 (+₹16,882)**, then 14:00 (+₹9,926); the 10am hour *loses* -₹7,363. `v5_classic` milder but same direction (post-1pm + ₹10,517, 55% green). **Conclusion: the 2nd half is where the profit is – the rule was removed; the flip is bidirectional and active all session.** (Caveat: the “168% of net realized after 1pm” exit-time figure is `TIME_EXIT`-inflated; the entry-time numbers are the clean measure.)

6.3 Validation 2 – “short=red, long=green” is REFUTED

P&L by direction × market-day-direction:

COHORT	LONG	SHORT
v5 · UP-day	+₹122,689 (71% green)	+₹11,747 (30%)
v5 · DOWN-day	+₹79,428 (56% green)	+₹17,584 (48%)
classic · UP-day	+₹59,170 (69%)	-₹3,711 (36%)
classic · DOWN-day	+₹19,787 (52%)	+₹33,457 (59%)

Two refutations: (1) **shorts profit on their own** (v5 net-positive in both regimes); (2) **“the other way around” works** – longs make money on DOWN days (+₹79,428, 56% green): strong stocks rise in a falling market. **The edge is stock-selection (long strong / short weak); the regime should tilt the ratio, not flip the book.** Re-arm winners on both sides – repeated-entry evidence: **COALINDIA shorted ×6 on 2026-04-10 = +₹23,197**; multi-entry shorts +₹40,714 vs single +₹21,429; multi-entry longs +₹280,808.

6.4 Validation 3 – tilt magnitude & trigger

Per-trade P&L bucketed by how red the day actually was:

SEVERITY (NIFTY O→C)	V5 LONG/TRADE	V5 SHORT/TRADE
UP (>+0.15%)	+203 (74%)	+18 (31%)
MILD-DOWN (−0.15 to −0.6%)	+123 (63%)	+11 (47%)
HARD-DOWN (< −0.6%)	+49 (45%)	+82 (53%)

Trigger correction: the short-tilt fires only on **HARD-DOWN (< −0.6%)** – mild-down days still favour longs for v5. **Magnitude:** the engine’s existing **BEAR 8L/12S** slot split is already in the data-supported zone (hard-down: v5 +₹1,378/day, classic +₹1,023/day). The linear ratio-sim shows more shorts (2L/18S) earns more on paper, but ⚠️ that overstates marginal shorts (best-scored-first), over-concentrates, ignores afternoon-reversal risk, and rests on only 7–8 hard-down days – **do not chase it**. Never zero longs (v5 longs earn +49/trade even hard-down). **So the fix is fast intraday activation of the existing 8/12, not a new ratio.**

6.5 Validation 4 – fixed ratio vs dynamic allocation

Does the engine adapt its mix to the tape? **No.** Short-share by day severity:

DAY	SHORT SHARE OF BOOK
UP (>+0.15%)	46%
FLAT	43%
MILD-DOWN	45%
HARD-DOWN (< −0.6%)	45%

The mix is flat ~45% whether the day is +0.5% green or −1% red. The BEAR split exists in code but rarely activates (slow daily regime) → a *de-facto* fixed ratio. **The principled design** (the user’s point): direction decided **per-stock by its own trend**, the long/short count **floats** from the opportunity set, and a **net-exposure risk cap** replaces the fixed ratio. This is dynamic market-neutral-with-tilt – what v5 was validated as.

Honest dependency: this only beats a fixed mix if the per-stock signal is predictive – and conviction→P&L could not be validated because score wasn’t stored on closed trades (§3.2, now fixed). The cross-sectional score is weakly predictive (winner≈loser; IC 0.006), so the driver should be **per-stock trend (VWAP/MA)**, not that score.

6.6 The 10% cap – clarification

The “10%” in code is `BASELINE_MAX_POSITION_PCT = 0.10` – **max 10% of pool capital per position (a size/concentration cap), not a loss cap** (loss is bounded tighter by the 1.5–2.25% stop; there’s also a 10% *monthly* pool-drawdown breaker). It covers the “max single-name” half of the risk cap but does **not** bound *net directional exposure* – so the dynamic design still needs a net-exposure limit on top. Necessary but not sufficient.

7. Features & Tools Built

7.1 Red-day watchdog – `scripts/red-day-watchdog.py`

Real-time (5-min) loss attribution: NIFTY direction · per-engine long-vs-short P&L split · regime-mismatch flag · the “if we were short-heavy” counterfactual · one-shot Telegram alert. Read-only, exits at 15:35. Built live during the red day; confirmed the diagnosis and ran all session.

7.2 Daily engine compare – `scripts/engine-compare.py` + `launchd`

A reusable script + `com.tradepilot.engine-compare.plist` (weekdays 15:40 IST) that Telegrams the full engine scorecard (net P&L, WR, L/S split) + a trailing cumulative. Now covers all 5 active engines. Verified live against real data.

7.3 Decision dashboard – `docs/decision-dashboard.html` + Flask `/decisions` + `pageswitch`

A single-source-of-truth strategy/roadmap dashboard, integrated into the live Flask app at `localhost:5050/decisions` (HTTP 200 verified), with a “Decisions” entry added to the global `static/pageswitch.js` nav so it appears on **every** page.

7.4 yfinance per-process cache fix – `data_nse.py` + `missed-opportunities-watchdog.py`

The 06-26 data-stall root fix (§3.1). Verified live.

7.5 Conviction logging – all engines

The “store everything” fix (§3.2). Verified; applies next launch.

7.6 `v5_flip` fast-regime-flip shadow – `scripts/v5_flip-paper-trade.py` + `wiring`

Same v5 code + an env-gated (`FAST_FLIP=1`) hook in the scan loop. Every **5 min** it re-checks the live NIFTY tape; on a **confirmed hard-down (< -0.6%, 2 consecutive reads)** it sets `state["regime"]="BEAR"` intraday (activating the existing 8/12 slot split via `pm.set_regime` + `rm.regime`); **bidirectional** (reverts to SIDEWAYS on a confirmed green reversal, ≥2 reads); keeps both legs; anti-whipsaw via confirmation + hysteresis. Env-gated so **live v5 is unaffected**. Wired into `launch-market.sh` , `crash-watchdog.sh` , and `engine-compare.py` (roster=5). **Verified:** all scripts compile, `bash -n` clean, roster=5 consistent in both files, env-override functional (`SCAN_INTERVAL_MIN=5` , `_fast_flip` callable), smoke gate PASS exit 0.

Honest scope: `v5_flip` is the *stepping-stone* (“fast activation of the existing 8/12 tilt”) – the data justified it. The *full* dynamic per-stock-trend allocation + net-exposure cap is deliberately **not** built; it’s gated on the conviction→P&L data now being collected.

8. Roadmap (all tracks)

TRACK	FOCUS	STATUS
A – Profitability	RC-1 long-only · RC-2 short-gate · RC-3 entry timing · RC-4 concentrate · (data-stall fixed, roster lean)	In progress; <code>v5_flip</code> + <code>v5_long</code> testing
B – Autonomy (T3→T4)	Meta-agent that proposes/auto-applies strategy-param changes	Future, post live track record
C – Competitive moat / product	Lead with autonomy (uncontested in India); build live execution layer	Designed
D – Compliance	Paper (zero) → live own-capital (light) → product (heavy)	Phased; D-0 now
E – Validation & risk	Keep alpha in execution; paper→live trust gate; audited track record; robustness	Ongoing
F – Research follow-ups	Numerai deep-dive · SEBI RA process · Jarvis Invest analog · Kite algo-TOS (before go-live) · Standard Signal EDGAR	Queued
G – Agentic platform	BYOB White-Box Agent ; G-0 personal-T4 (9-role DAG + meta-agent + train/eval-namespace) → G-1 invite-BYOB → G-2 beta (RA licence) → G-3 marketplace	Roadmapped; gated on Track A

Sequencing rule: prove profitability → go live cheaply (own capital) → productize into the agentic platform *only if chosen*. The agentic-DAG architecture (G-0) and the overfitting fix (Track A) are the **same** engineering investment – platform and profit converge.

9. Learnings Index (DevPilot store)

~20 learnings persisted this arc (`dp lrc recall "tradepilot <topic>"`): edge-source · over-trading · short-book (decisive) · `v5_classic` dominance · OLD-5tree skeptic · data-stall bug · competitive landscape · SEBI compliance · strategic position (T3) · 3-pillar strategy · red-day live finding · fast-flip design · 2nd-half validation · direction-vs-regime · tilt-magnitude · dynamic-allocation · conviction-logging + `v5_flip` build · session wrap-up.

10. Open Questions & Next Steps

- Does the morning loss recover by EOD on down days?** Inferred from aggregates (longs net-positive on down days) – worth a direct intraday-P&L-path check.
- Conviction→P&L** – answerable in ~5 sessions now that score is logged; settles the dynamic-allocation design.
- Purged folds vs leaky K-fold** in `backtest-honest-fills.py` – determines whether the 5-tree “win” is real before shrinking the live model.
- `v5_flip` ’s first red day** – does fast-activating the 8/12 tilt cut red-day losses *without* false-triggering on green days? (The decisive live test.)
- Reconcile the two long-only experiments** (`v5_long` vs the pre-existing `tradepilot-v5-longonly-ab`).
- Net-exposure cap** – the missing risk control for the dynamic phase.
- Kite Connect algo-TOS** – does own-account automation below 10 OPS need separate approval? (Gate before any go-live.)

End of report. Companion docs: [ROOT_CAUSE_ANALYSIS_2026-06-24](#) , [_II_2026-06-26](#) , [COMPETITIVE_LANDSCAPE_2026-06-28](#) ,
[STRATEGY_RESEARCH_2026-06-30](#) , [DESIGN_fast-regime-flip_2026-06-30](#) , [ROADMAP_2026-06-28](#) , [SESSION_CHANGELOG_2026-06](#) .